

What the code computes, in the notation of the paper

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Equation numbers refer to arXiv:2608.16107v1. All quantities below are exactly those of the paper; only one auxiliary function, $H_i(E_\nu)$ in Eq. (N1), is introduced because it is needed to state precisely what the code does. Throughout, the split energy is 2 MeV (as in the paper and in the code).

1. Definitions taken from the paper

- $\phi(E_\nu)$: differential flux (Appendix B); $\Phi(E_\nu) = \int_{E_\nu}^{\infty} d\tilde{E}_\nu \phi(\tilde{E}_\nu)$, Eq. (2.3), so $\phi = -d\Phi/dE_\nu$.
- $\partial\mathcal{H}/\partial E'(E_\nu, E')$: response function acting on ϕ , Eqs. (4.4)–(4.5), containing $\epsilon(E')$, the resolution $G(E', E_R)$ and $\partial\sigma/\partial E_R$. $\partial\mathcal{R}/\partial E' \equiv \partial_{E_\nu}(\partial\mathcal{H}/\partial E')$, Eq. (4.8), acting on Φ .
- $R_i(E_\nu) \equiv R_{[E'_i, E'_{i+1}]}(E_\nu) = \int_{E'_i}^{E'_{i+1}} dE' \partial\mathcal{R}/\partial E'$, Eq. (4.12), in the notation of Fig. 3; $R_{ij} = \int_{E_\nu^j}^{E_\nu^{j+1}} dE_\nu R_i(E_\nu)$, Eq. (5.3).
- Ansatz: $\Phi(E_\nu) = \Phi_j$ on the j th of the N_{int} intervals of $[E_\nu^{\text{min}}, 2 \text{ MeV}]$, Eq. (5.1).

The bin-integrated response acting on ϕ (the same integral as Eq. (4.12) but of $\partial\mathcal{H}/\partial E'$) is

$$H_i(E_\nu) \equiv \int_{E'_i}^{E'_{i+1}} dE' \frac{\partial\mathcal{H}}{\partial E'}(E_\nu, E'), \quad R_i(E_\nu) = \frac{dH_i}{dE_\nu}, \quad H_i(0) = 0. \quad (\text{N1})$$

$H_i(0) = 0$ for the reason given below Eq. (4.7) ($E_R^{\text{max}} \rightarrow 0$). Hence $H_i(E_\nu) = \int_0^{E_\nu} dE R_i(E)$ and, with the intervals of Eq. (5.3),

$$H_i(2 \text{ MeV}) = \int_{E_\nu^{\text{min}}}^{2 \text{ MeV}} dE_\nu R_i(E_\nu) = \sum_{j=1}^{N_{\text{int}}} R_{ij}. \quad (\text{N2})$$

This answers the question about the boundary term: the function multiplying $\Phi(2 \text{ MeV})$ is $\partial\mathcal{H}/\partial E'$ at $E_\nu = 2 \text{ MeV}$ integrated over the bin, i.e. $H_i(2 \text{ MeV})$, and by (N2) it is just the row sum of the response matrix.

2. What the code computes

Rates per unit mass ($\text{kg}^{-1}\text{yr}^{-1}$); the exposure $\mathcal{E}$ multiplies everything as in Eqs. (4.10), (5.4).

$$N_i^{\text{tot}} = \int_0^\infty dE_\nu \phi(E_\nu) H_i(E_\nu) \quad \text{Eqs. (4.6),(4.10); black line of Fig. 2,} \quad (\text{N3})$$

$$N_i^{<2} = \int_0^{2 \text{ MeV}} dE_\nu \phi(E_\nu) H_i(E_\nu) \quad \text{Eq. (4.6), upper limit 2 MeV; cyan line of Fig. 2,} \quad (\text{N4})$$

$$h_i^{\text{code}} = N_i^{\text{tot}} - N_i^{<2} = \int_{2 \text{ MeV}}^\infty dE_\nu \phi(E_\nu) H_i(E_\nu) \quad \text{green line of Fig. 2 (= } h_i \text{ of Ref. [6]).} \quad (\text{N5})$$

N_i^{tot} agrees with Fig. 3 of Ref. [6]; h_i^{code} is what the text calls the contribution of neutrinos with $E_\nu > 2 \text{ MeV}$. The mock data are $O_i = N_i^{\text{tot}} + b_i$ (Poisson fluctuated per pseudo-experiment). The fit minimises exactly Eq. (5.7),

$$\chi^2 = \sum_{i=1}^d \frac{[\mathcal{E}(\sum_j R_{ij} x_j + h_i^{\text{code}} + b_i) - \mathcal{E}O_i]^2}{\mathcal{E}O_i}, \quad x_j \geq 0, \quad x_{j+1} \leq x_j, \quad (\text{N6})$$

with the constraints of Eq. (5.8), the same R_{ij} , the same N_{int} , the same solver and the same Monte-Carlo band construction as in Appendix C. The only difference from Eq. (5.4) is that h_i is h_i^{code} of (N5), and the fitted parameters are called x_j here because, as shown next, they are not Φ_j .

3. What the fitted parameters are

Apply the integration by parts of Eq. (4.7) to $N_i^{<2}$, i.e. with the upper limit 2 MeV instead of ∞ :

$$N_i^{<2} = \left[-\Phi(E_\nu) H_i(E_\nu) \right]_0^{2 \text{ MeV}} + \int_0^{2 \text{ MeV}} dE_\nu \Phi(E_\nu) R_i(E_\nu) = -\Phi(2 \text{ MeV}) H_i(2 \text{ MeV}) + \sum_j R_{ij} \Phi_j. \quad (\text{N7})$$

The lower boundary term vanishes as in the paper. The upper one does not: $\Phi(2 \text{ MeV}) \neq 0$ (Fig. 1). Using (N2),

$$\boxed{N_i^{<2} = \sum_j R_{ij} (\Phi_j - \Phi(2 \text{ MeV}))} \implies x_j = \Delta\Phi_j \equiv \Phi_j - \Phi(2 \text{ MeV}). \quad (\text{N8})$$

Equivalently, defining $h_i^{\text{paper}} \equiv \int_{2 \text{ MeV}}^\infty dE_\nu \Phi(E_\nu) R_i(E_\nu)$ (the Φ -weighted response above 2 MeV),

$$h_i^{\text{code}} = h_i^{\text{paper}} + \Phi(2 \text{ MeV}) \sum_j R_{ij}, \quad N_i^{\text{tot}} = \underbrace{\sum_j R_{ij} \Phi_j}_{\text{Eq. (5.4)}} + \underbrace{\sum_j R_{ij} \Delta\Phi_j}_{\text{code}} + h_i^{\text{code}}. \quad (\text{N9})$$

Both decompositions are exact and give the same total; they differ in whether the term $\Phi(2 \text{ MeV}) \sum_j R_{ij}$ is counted as signal or as “high-energy” contribution. Eq. (5.4) with Φ_j requires h_i^{paper} ; the code (and the definition of h_i in the text and in Fig. 2 as the rate from neutrinos above 2 MeV) uses h_i^{code} , which forces the fitted parameters to be $\Delta\Phi_j$. I have checked (N7)–(N9) numerically with the response table of the code (better than 1%).

4. Why $\Delta\Phi$ appeared; is it better?

It was not a choice and it is not better. I computed h_i as the physical rate from neutrinos above 2 MeV, following Ref. [6] and Fig. 2, and this automatically turns the fitted parameters into $\Delta\Phi_j$. There is no advantage: the total rate, χ^2 and constraints are the same, but what comes out is $\Phi(E_\nu) - \Phi(2 \text{ MeV})$, which is why the best-fit and the theory curves in Fig. 7 lie below Fig. 1 by $\Phi(2 \text{ MeV}) \simeq 4.6 \times 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$ (24% of $\Phi(0.18 \text{ MeV})$; solid curve of Fig. 1). The paper is written for Φ , and that is what we want.

5. Size of the term (1 eV threshold; $\text{kg}^{-1} \text{ yr}^{-1}$ per bin; solid flux of Fig. 1)

bin [eV]	$N_i^{<2} = \sum_j R_{ij} \Delta\Phi_j$	$\Phi(2 \text{ MeV}) \sum_j R_{ij}$	$\sum_j R_{ij} \Phi_j$	h_i^{paper}	h_i^{code}	N_i^{tot}
1–3	232.5	96.9	329.4	0.8	97.7	330.2
11–13	121.9	92.3	214.2	4.9	97.3	219.2
21–23	76.1	83.7	159.8	9.1	92.8	168.9
41–43	33.0	66.5	99.5	17.3	83.8	116.8
51–56	52.0	141.5	193.5	55.1	196.6	248.6
81–120	32.7	316.7	349.3	806.5	1123.1	1155.8

The term $\Phi(2 \text{ MeV}) \sum_j R_{ij}$ is comparable to or larger than $N_i^{<2}$ in every bin, so the difference between fitting Φ_j and $\Delta\Phi_j$ is large.

6. What has to change

Two self-consistent options:

- (a) **Fit Φ , as written in Eq. (5.4).** Signal = $\sum_j R_{ij}\Phi_j = N_i^{<2} + \Phi(2\text{ MeV}) \sum_j R_{ij}$, background = h_i^{paper} . All fits and bands must be redone (both thresholds); Fig. 2 (green and cyan lines) and the sentence defining h_i must be changed (h_i is then the Φ -weighted response above 2 MeV, not the number of events from neutrinos above 2 MeV; at low E' it is tiny, see the table). The comparison with Ref. [6] must be redone.
- (b) **Keep the numerics, rewrite in $\Delta\Phi$.** Eqs. (5.1), (5.2), (5.4), (5.7), (5.8) and the vertical axes of the flux figures in terms of $\Delta\Phi_j = \Phi_j - \Phi(2\text{ MeV})$ (constraints $\Delta\Phi_j \geq 0$, non-increasing, hold as well); h_i , Fig. 2 and all fits unchanged; the theory curves shifted by $\Phi(2\text{ MeV})$.

Since the paper is written for Φ , I propose (a). I can start immediately; the code changes are only the replacement of the signal and h_i vectors, but the confidence bands have to be recomputed.

Note on inputs: the code implements the resolution as a box of half-width 1 eV (the paper states this is equivalent to the Gaussian of Eq. (3.3)) and $\epsilon(E') = \theta(E' - E'_{\text{thr}})$; the numbers in the table use the response table of the code and the solid flux of Fig. 1.